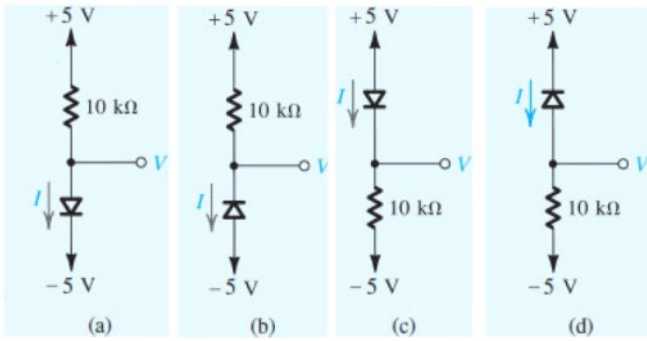
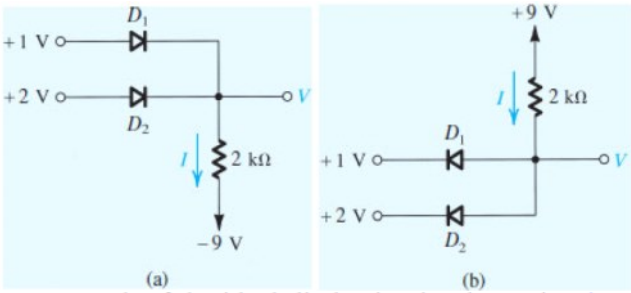
Questions

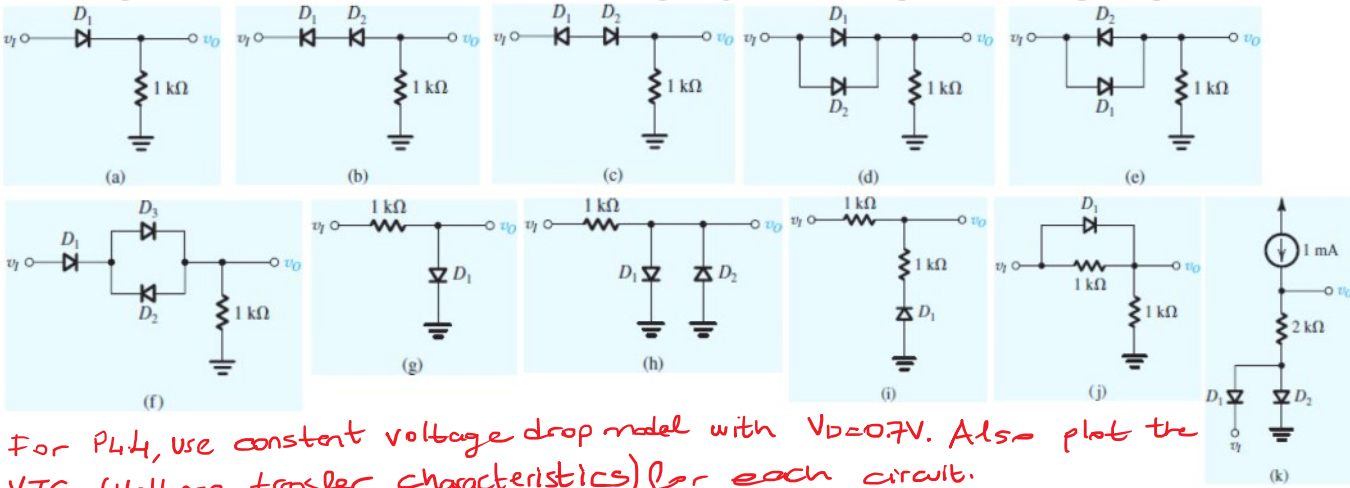
4.2 For the circuits shown in Fig. P4.2(a), (b), (c), and (d) using ideal diodes, find the values of the voltages and currents indicated.



4.3 For the circuits shown in Fig. P4.3(a) and (b) using ideal diodes, find the values of the labeled voltages and currents.



4.4 In each of the [redacted]-diode circuits shown in Fig. P4.4(a), (b), (c), (d), (e), (f), (g), (h), (i), (j), and (k), v_I is a 1-kHz, 5-V peak sine wave. Sketch the waveform resulting at v_O. What are its positive and negative peak values?



For P4.4, use constant voltage drop model with V_D=0.7V. Also plot the VTC (voltage transfer characteristics) for each circuit.

4.18 Calculate the value of the thermal voltage, V_T, at −55°C, 0°C, +40°C, and +125°C. At what temperature is V_T exactly 25 mV?

4.20 A diode for which the forward voltage drop is 0.7 V at 1.0 mA is operated at 0.6 V. What is the value of the current?